Product Strategy Document

PharmaIntel India

Universal Multilingual Indian Pharmaceutical Intelligence Platform

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Document Type: Product Strategy & Design Overview

Version: 2.0 — Final

Date: March 2026

1. Executive Summary

PharmaIntel India is a comprehensive offline-first pharmaceutical intelligence platform designed for India's diverse, multilingual healthcare ecosystem. It delivers verified drug information, AI-powered pill identification, drug interaction checking, and market intelligence across Android, iOS, Web (PWA), Windows, and macOS — from a single codebase — supporting 13 Indian languages without requiring an internet connection.

The platform addresses a fundamental information asymmetry in Indian healthcare: patients and frontline health workers in Tier-2 and Tier-3 cities often cannot access reliable, language-appropriate drug information. PharmaIntel India closes this gap through a combination of a locally stored medicine database (~31 MB compressed), on-device artificial intelligence, and strict regulatory alignment with CDSCO, NLEM 2022, and the Digital Personal Data Protection Act, 2023.

5	13	50,000+	<100ms	<1s
Platforms	Languages	Medicines	Search	App Launch

2. Problem Statement

India's pharmaceutical market — ranked third globally by volume — serves a population of 1.4 billion across 22 official languages and vastly unequal connectivity conditions. Despite this scale, the tools available for drug information access are fragmented, largely English-only, and dependent on continuous internet connectivity. Several structural gaps persist:

Language exclusion: Most drug information platforms operate exclusively in English, excluding the majority of India's rural and semi-urban population.

Connectivity dependency: Cloud-dependent applications fail in low-signal environments where health workers and pharmacists operate daily.

Affordability blind spots: Patients are rarely shown cheaper generic alternatives or NLEM price-controlled equivalents at the point of dispensing.

Interaction risk: Drug interaction checking requires either specialist software or physician consultation — neither accessible in rural settings.

Regulatory opacity: Schedule classification (OTC vs. prescription vs. controlled) is not communicated clearly to pharmacists and consumers.

3. Target Users

User Segment	Primary Need	Key Features Used
Retail Pharmacists	Quickly verify drug information, check interactions, find substitutes	Medicine Search, Interaction Checker, Substitute Finder
Hospital Pharmacists	Clinical drug data with source citations for dispensing decisions	Medicine Detail, Disease Mapping, Knowledge Graph

User Segment	Primary Need	Key Features Used
Medical Representatives	Market intelligence, therapy-area analytics for field work	Market Dashboard, Therapy Analytics
MBBS / BAMS Students	Reference tool for pharmacology, drug schedules, side effects	Medicine Detail, Side Effects, Disease Browser
Health & Wellness Consumers	Identify pills, check basic interactions, understand prescriptions	Pill Identifier, Symptom Checker, Search
Pharma Brand Managers	Competitive landscape, market share, pricing trends	Market Intelligence Dashboard, Report Upload

4. Core Product Features

4.1 Multilingual Medicine Search

The medicine search engine supports fuzzy matching, phonetic transliteration, and cross-script queries across all 13 languages. A user searching for '[redacted]' (Metformin in Hindi) or its transliteration in any supported script retrieves the same comprehensive drug profile. Search latency is guaranteed below 100 milliseconds via an on-device FlexSearch index operating in a dedicated Web Worker thread, entirely offline.

Each medicine profile includes: brand and generic name, composition, schedule classification (colour-coded: green for OTC, amber for H, orange for H1, red for X), MRP, NLEM 2022 price ceiling where applicable, manufacturer, therapeutic class, side effects (grouped by MedDRA frequency), and source citations for all clinical data.

4.2 Affordable Substitute Finder

For any branded medicine, the platform identifies therapeutically equivalent alternatives at lower price points — including NLEM-listed generics. Savings percentages are calculated and displayed at the point of lookup, enabling pharmacists and patients to make informed substitution decisions within the bounds of the Drugs & Cosmetics Act, 1940.

4.3 Drug Interaction Checker

Users can select two or more medicines simultaneously and receive severity-coded interaction assessments — derived from the WHO Model Formulary, PubMed literature, and Indian Pharmacopoeia. Results are classified as Major, Moderate, or Minor, with mechanism of interaction, clinical effect, and recommended management for each pair. Every output includes a mandatory disclaimer and source reference (DOI/PMID).

4.4 AI Pill Identifier

A MobileNetV3-Small model, quantised to INT8 (~2 MB) and running entirely on-device via ONNX Runtime Web, classifies a captured or uploaded pill image into the top 3–5 candidate medicines with associated confidence scores. When model confidence falls below 60%, the system falls back to a rule-based matcher using shape, colour, imprint, and scoring attributes. No image data leaves the device. Inference completes in under two seconds on mid-range Android hardware.

4.5 Bayesian Symptom Checker

A Naive Bayes probabilistic engine accepts structured symptom inputs (curated list, not free text) and returns ranked differential diagnoses with posterior probability scores sourced from peer-reviewed epidemiological literature. The feature is gated behind an unavoidable disclaimer screen — it does not constitute a clinical diagnosis and is positioned explicitly as an educational reference tool.

4.6 Market Intelligence Dashboard

Pharmaceutical professionals can upload proprietary market reports (CSV/XLSX) or use curated sample datasets to generate therapy-area analytics, company market share charts, pricing trend visualisations, and competitive landscape summaries. Charts are exportable as PNG or PDF for field presentations and internal briefings.

4.7 Knowledge Graph Explorer

An interactive graph visualises relationships between medicines, active compositions, diseases, manufacturers, and therapy areas. Edge types include TREATS, CONTAINS, MANUFACTURED_BY, INTERACTS_WITH, SUBSTITUTE_OF, and CONTRAINDICATED_IN. This provides a navigable, citation-backed view of the pharmaceutical ecosystem useful for academic research, pharmacovigilance, and brand strategy.

5. Technical Architecture

The application is built as a single React 19 + TypeScript codebase, deployed across five platforms using Capacitor (Android/iOS) and Electron (Windows/macOS) shell wrappers. All business logic, AI inference, and pharmaceutical data reside on-device, with an optional cloud backend handling only delta synchronisation and analytics aggregation.

Layer	Components	Technology
Presentation	Web PWA, Android, iOS, Windows, macOS shells	React 19 + Capacitor / Electron
Application	Medicine Search, Interaction Checker, Pill ID, Symptom Checker, Market Dashboard, Knowledge Graph	TypeScript feature modules (Feature-Sliced Design)
Data Access	Unified data access layer with platform abstraction	Dexie.js / SQLite + TanStack Query
Storage	Local DB, FlexSearch index, ONNX model cache, language packs	IndexedDB / SQLite / localStorage
Cloud (Optional)	Delta sync API, report storage, anonymised analytics	Hono.js + PostgreSQL + Cloudflare R2

Offline-First Design

The application ships with a compressed medicine database (~31 MB gzipped), decompressed into local storage on first launch via a background worker thread. Priority loading ensures the 5,000 most commonly dispensed medicines are searchable within two to three seconds of first launch, before full hydration completes. Subsequent launches require no network access whatsoever for core features.

Platform-specific storage engines are abstracted behind a unified data access interface: Dexie.js (IndexedDB) for Web PWA, @capacitor-community/sqlite with FTS5 and SQLCipher encryption for mobile, and better-sqlite3 for Electron desktop. This abstraction allows feature code to remain platform-agnostic.

Performance Targets

Metric	Target	Method
App launch time	< 1 second	Bundled seed + priority loading of top 5,000 medicines
Search latency	< 100 ms	On-device FlexSearch index in dedicated Web Worker
Pill ID inference	< 2 seconds	INT8-quantised MobileNetV3 (~2 MB) via ONNX Runtime Web
Initial JS bundle	< 150 KB gzipped	Route-based code splitting; ONNX runtime loaded on demand
Offline DB size	~31 MB compressed	All medicines, interactions, side effects, disease mappings
Lighthouse score	≥ 95	Automated Lighthouse CI on every production build

6. Compliance and Safety Framework

PharmaIntel India is an information tool, not a diagnostic or prescription system. This distinction is enforced architecturally and reflected in every user-facing surface.

Regulation	Requirement	Implementation
CDSCO Schedule Classification	Correctly classify OTC / H / H1 / X	Schedule field validated at ingestion against CDSCO master list; colour-coded in UI
NLEM 2022	Highlight essential medicines and price ceilings	nlemListed flag + nlemPrice field; NLEM badge on medicine cards
Indian Pharmacopoeia	Standard nomenclature for all salts and generics	All generic names cross-referenced with IP at data ingestion
Drugs & Cosmetics Act 1940	No Schedule H/X promotion without prescription gate	UI warnings on all controlled substances; no promotional language permitted
DPDPA 2023	No personal health data on servers	All AI inference on-device; clearAllUserData() flow; no personal data transmitted
IT Act 2000 §43A	Reasonable data security	SQLCipher (256-bit AES) on mobile; TLS 1.3 with certificate pinning; no eval()

All pharmaceutical data is validated against at least one of: CDSCO approval records, NLEM 2022 gazette notification, Indian Pharmacopoeia, WHO Model Formulary, NIH DailyMed, FDA Orange Book, or peer-reviewed publications (PubMed, JAMA, Lancet). Data points that cannot be confirmed from an approved source are excluded or marked explicitly as unverified.

7. Platform and Deployment Strategy

A Web-first (PWA) release strategy is recommended: the PWA reaches users immediately without App Store approval cycles, enables rapid iteration, and serves as the reference implementation for the mobile and desktop shells. Android and iOS builds follow using Capacitor, with Windows and macOS Electron builds completing the platform matrix.

Platform	Build Tool	Output	Distribution Channel
Web (PWA)	Vite	Static HTML/JS/CSS	Cloudflare Pages — global CDN, auto-deploy from main
Android	Capacitor + Gradle	AAB	Google Play Store
iOS	Capacitor + Xcode	IPA	Apple App Store / TestFlight
Windows	Electron Forge + NSIS	.exe installer	Direct download / Microsoft Store
macOS	Electron Forge	.dmg / .app	Direct download / Mac App Store

CI/CD is automated via GitHub Actions: every commit to the main branch triggers linting, TypeScript type-checking, unit tests (Vitest), component tests (React Testing Library), E2E tests (Playwright across Chrome, Safari, Firefox), Lighthouse CI validation, and production builds for all five platforms.

8. Development Roadmap

The platform is designed for development by a single developer over 14–16 weeks, structured as seven sequential phases. Each phase produces a testable, deployable increment.

Phase	Duration	Key Deliverables
1 — Foundation	Weeks 1–2	React + TypeScript scaffold, Tailwind design tokens, shared UI library, routing, Zustand stores, i18next (EN + HI), Dexie.js schema, platform abstraction layer
2 — Core Medicine Features	Weeks 3–4	Medicine search (FlexSearch + transliteration), Medicine Detail page, Substitute Finder, Side Effects panel, Schedule/NLEM badges, DPDPA user data flows
3 — Safety & Clinical Tools	Weeks 5–6	Drug Interaction Checker, Bayesian Symptom Checker, Disease-Medicine Mapping browser, source citations on all clinical outputs, error handling and sync retry logic
4 — AI Features	Weeks 7–9	ONNX Runtime Web integration, camera capture + preprocessing, MobileNetV3 pill identification, rule-based fallback matcher, Knowledge Graph engine + explorer
5 — Multilingual Expansion	Weeks 10–11	All 13 language files, Indic font loading, RTL layout for Urdu, locale-aware number formatting, cross-script search indexing, visual regression in all 13 languages

Phase	Duration	Key Deliverables
6 — Market Intelligence	Weeks 12–13	Market Dashboard, Recharts visualisations (Pie/Line/Bar/Radar), CSV/XLSX import, market share and pricing charts, therapy-area analytics, chart export to PNG/PDF
7 — Multi-Platform & Production	Weeks 14–16	Capacitor (Android/iOS) and Electron (Windows/macOS) configuration, Workbox Service Worker, Lighthouse CI ≥95, WCAG 2.1 AA audit, Playwright E2E, App Store submission

9. Data Strategy

The quality and regulatory provenance of the underlying pharmaceutical dataset is the most critical determinant of product credibility. All data must be traceable to an approved source before inclusion. The following datasets are required at launch:

Dataset	Minimum Records	Primary Sources	Launch Priority
Medicine Master Database	50,000+	CDSCO approval records, NLEM 2022 gazette	Critical
Drug Interaction Matrix	10,000+ pairs	WHO Model Formulary, PubMed, Indian Pharmacopoeia	Critical
Side Effects Database	100,000+ records	Indian Pharmacopoeia, NIH DailyMed	Critical
NLEM 2022 Price List	384 medicines	Official NLEM 2022 gazette notification	Critical
Disease-Medicine Mapping	2,000+ diseases	WHO treatment guidelines, Indian clinical practice guidelines	High
Pill Image Dataset	5,000+ (50 per class)	Photographed or verified labelled image databases	High
Symptom-Disease Probabilities	500+ symptom-disease pairs	Epidemiological literature, PubMed DOI/PMID	High
Market Intelligence Reports	Recent data	IQVIA, Pharmatrac, or user-uploaded proprietary reports	Medium

10. Risk Management

Risk	Severity	Mitigation
Insufficient pill image training data	High	Launch with rule-based matcher (shape + colour + imprint). ML model added when ≥50 labelled images per class are available.
App Store rejection for medical content	High	Prominent disclaimers on all clinical screens. Avoid 'diagnosis' or 'prescription' in store listing and marketing copy. Position as reference tool.
DPDPA compliance gap	High	On-device processing for all AI and clinical data. No personal health data on servers. Full clearAllUserData() deletion flow accessible from Settings.
Medical translation quality	High	Professional translators for disclaimer and medical glossary files. Machine translation restricted to non-clinical UI strings after human review.
ONNX inference speed on low-end devices	Medium	Ship INT8-quantised model (~2 MB). Offload inference to Web Worker. Display rule-based fallback if inference exceeds five seconds.

Risk	Severity	Mitigation
IndexedDB quota on Safari	Medium	Monitor storage quota with navigator.storage.estimate(). Implement data pruning. Test on Safari 17+ which lifted prior storage restrictions.
Regulatory data accuracy drift	Medium	Quarterly data refresh cycle. Version-tagged datasets with update notifications delivered to users via delta sync.

11. Success Metrics

Technical Metrics

Search latency below 100 ms on all platforms under test conditions.

App launch time below one second after initial seed is loaded.

Lighthouse performance, accessibility, and best-practices scores all ≥ 95.

Zero critical or high-severity Sentry errors in production for 30 consecutive days post-launch.

Playwright E2E test suite passing on Chrome, Safari, and Firefox before each release.

User and Product Metrics

Monthly active users across all platforms (target: 10,000 within six months of launch).

Search-to-detail conversion rate — indicating users find the medicines they look for.

Drug interaction checker usage per session — a proxy for clinical utility.

Language distribution of active users — tracking reach beyond English-speaking segments.

Substitute Finder adoption rate and average savings displayed.

Market Dashboard report upload frequency among professional users.

12. Conclusion

PharmaIntel India addresses a real and significant gap in India's healthcare information infrastructure. By combining an offline-first architecture, comprehensive multilingual support, on-device AI, and rigorous regulatory compliance, the platform serves a broad spectrum of users — from rural pharmacists who need instant drug information without internet access, to pharmaceutical brand managers who need therapy-area analytics for field briefings.

The technical architecture is production-ready and fully specified. The development roadmap is structured for incremental delivery, with a publicly accessible Web PWA available at the end of Phase 2 — giving early users access to the core medicine search and substitute finder features before full platform parity is achieved.

The primary determinant of long-term product success is the quality and currency of the underlying pharmaceutical dataset. Investment in data quality — verified against CDSCO, NLEM, Indian Pharmacopoeia, and peer-reviewed literature — is therefore the highest-priority non-technical action before development commences.